

DAY 2 OF DAILY DEEP-DIVE SERIES

Does AI Actually read words?

The Real unit of AI Language

The Token

Text input:

"Artificial intelligence"

Art

if

icial

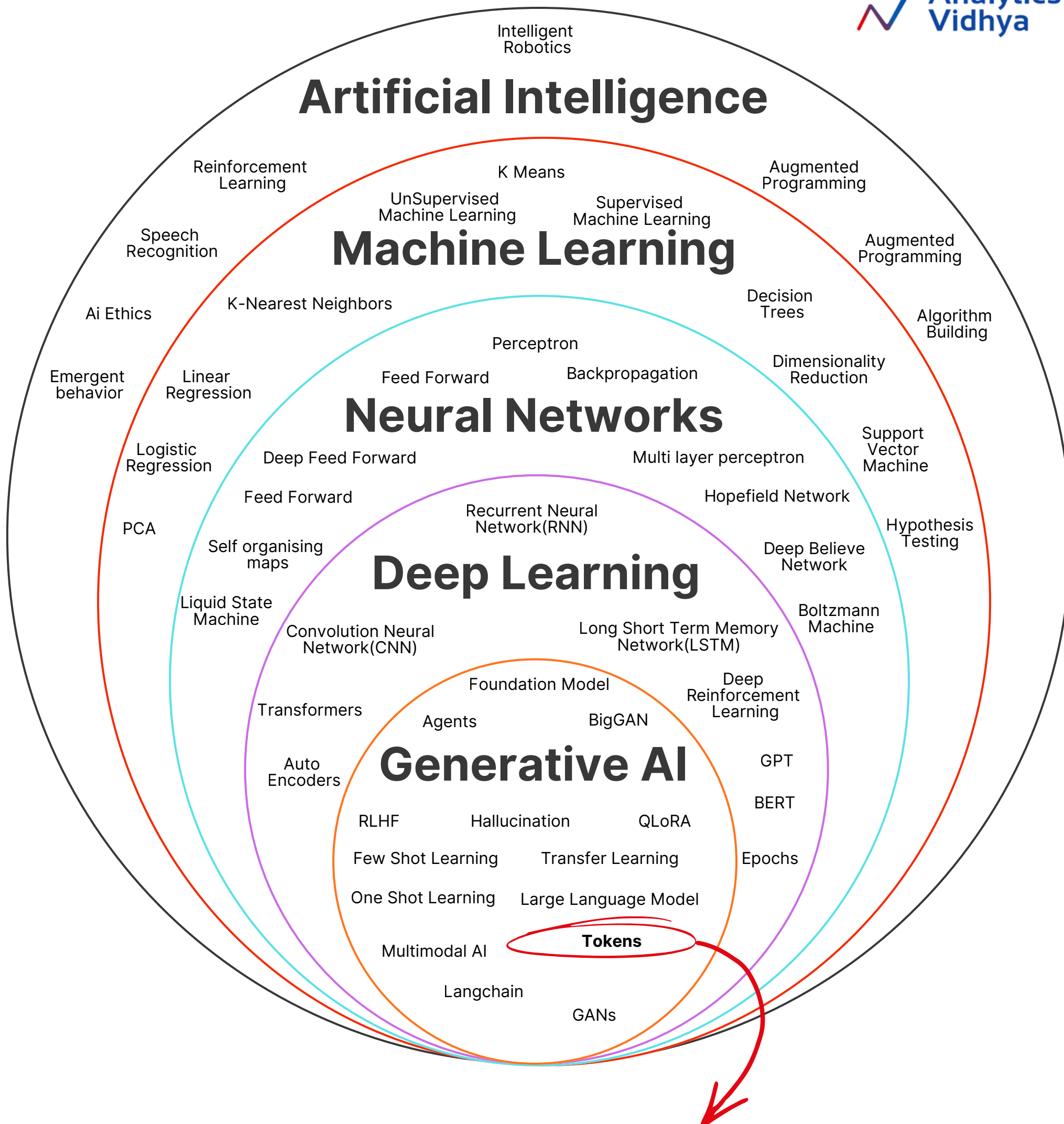
intell

igence

= 5 tokens

Not 2 words. Not 20 characters. 5 tokens.

Swipe to understand what tokens really are



A token is the smallest chunk of text a language model processes.

What exactly **is a token**?

A token is the smallest chunk of text a language model processes.

EXAMPLES

"cat"

cat = 1 token

"running"

run **ning** = 2 tokens

"Artificial"

Art **if** **icial** = 3 tokens

"2024"

2024 = 1 token

The golden rule

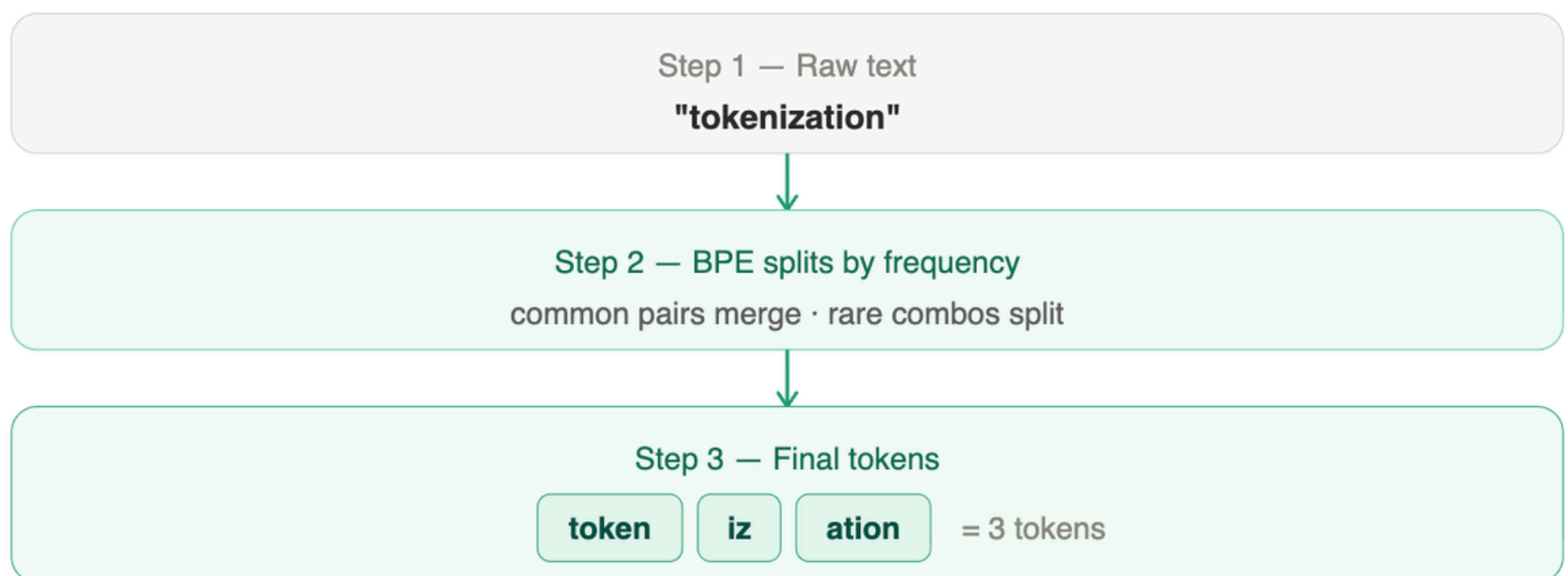
1 token $\approx \frac{3}{4}$ of an English word

100 words $\approx$ 130 tokens

How does text **become tokens**?

An algorithm called **Byte Pair Encoding (BPE)** learns the most efficient vocabulary from billions of words of training data

THE PROCESS



Why it matters for languages

English: efficient - more training data available

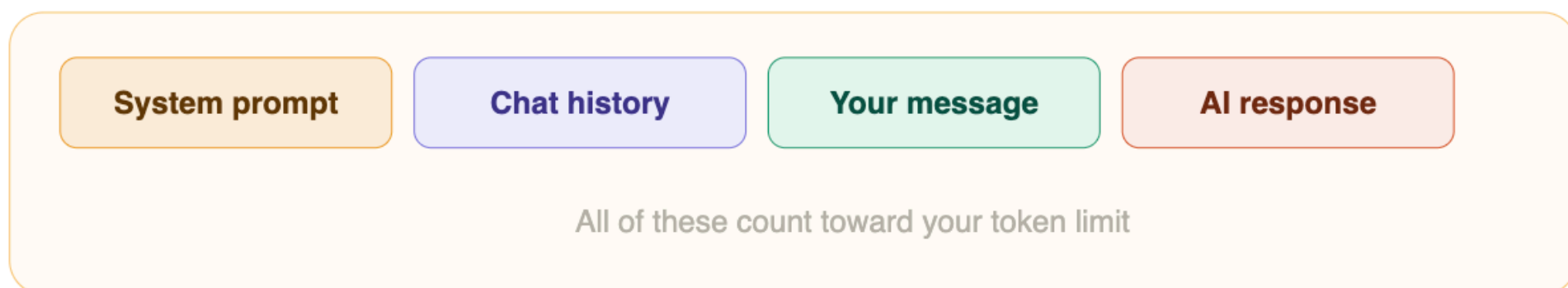
Hindi / Swahili: more tokens per word → costs more

The context window

Every token you send + every token the model generates shares one fixed pool of space.

Exceed it - and the model forgets.

WHAT LIVES INSIDE THE WINDOW



CONTEXT LIMITS BY MODEL

Claude 200K tokens	GPT-4o 128K tokens	Early GPT-3 4K tokens
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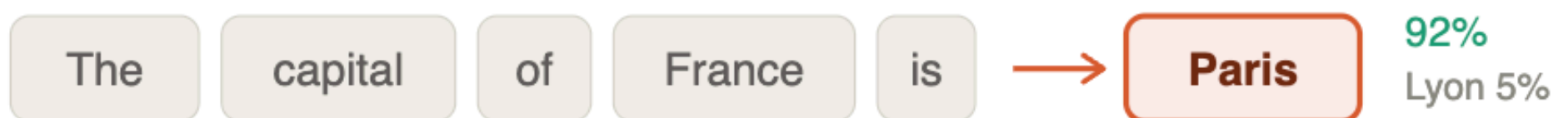
200K tokens ≈ a full 400-page book
read entirely, in one conversation

AI doesn't "think" - it predicts

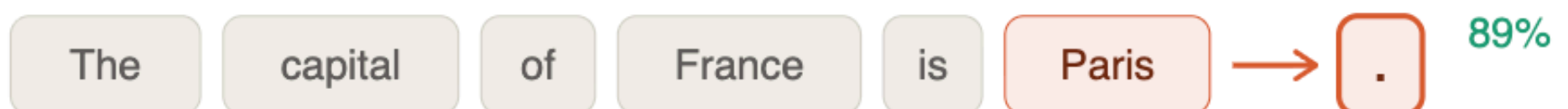
Every response is generated one token at a time. The model picks the most probable next token. Then repeats - until the response is complete

Watch it happen

Prompt tokens:



Next round — "Paris" is now part of the input:



This is why hallucinations happen.

The model is not retrieving facts from memory. It is completing a sequence based on statistical patterns in training data.

Probable ≠ Accurate